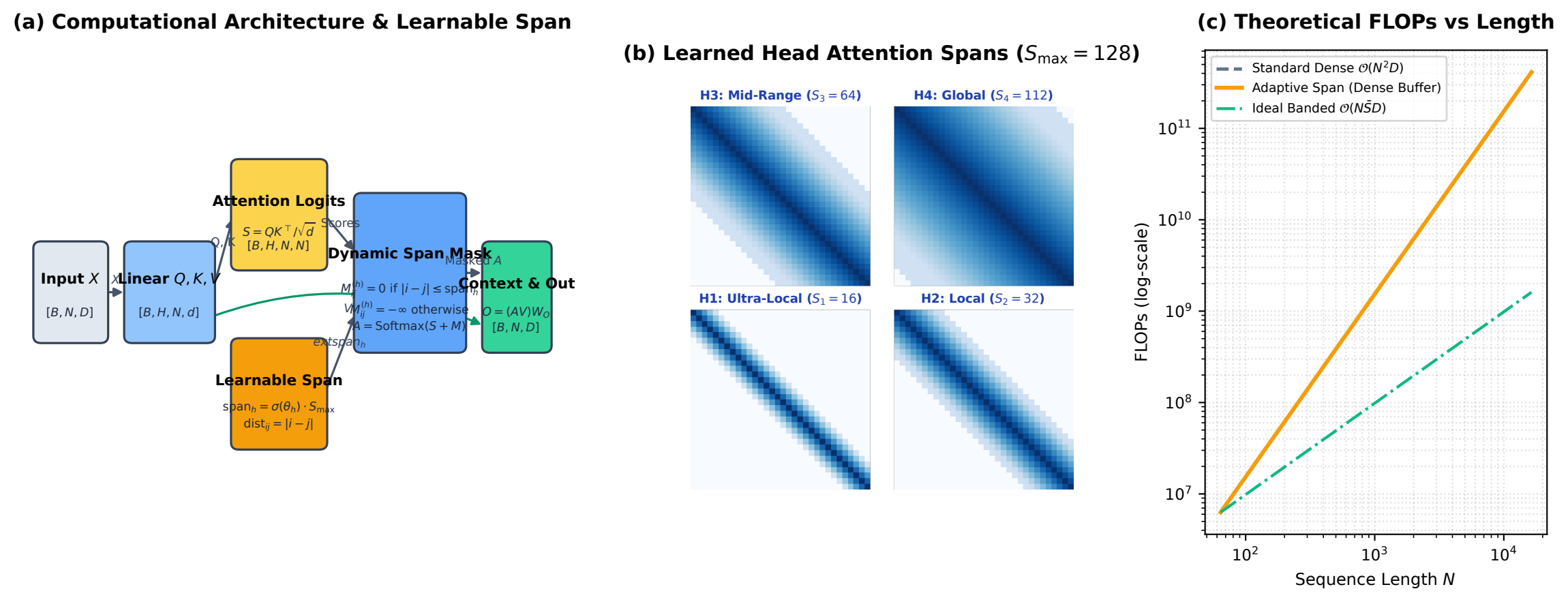

(a) Computational Architecture & Learnable Span
Input X
$[B, N, D]$
Linear Q, K, V
$[B, H, N, d]$
X
Q, K
Attention Logits
$S = QK^\top/\sqrt{d}$
$[B, H, N, N]$
Scores
Dynamic Span Mask
$M_{ij}^{(h)} = 0$ if $|i-j| \le \text{span}_h$
$M_{ij}^{(h)} = -\infty$ otherwise
$A = \text{Softmax}(S + M)$
Masked A
Context & Out
$O = (AV)W_O$
$[B, N, D]$
V
extspan_h
Learnable Span
$\text{span}_h = \sigma(\theta_h) \cdot S_{\max}$
$\text{dist}_{ij} = |i-j|$
(b) Learned Head Attention Spans ($S_{\max} = 128$)
H3: Mid-Range ($S_3 = 64$)
H4: Global ($S_4 = 112$)
H1: Ultra-Local ($S_1 = 16$)
H2: Local ($S_2 = 32$)
(c) Theoretical FLOPs vs Length
Standard Dense $\mathcal{O}(N^2D)$
Adaptive Span (Dense Buffer)
Ideal Banded $\mathcal{O}(N\bar{S}D)$
FLOPs (log-scale)
10^{11}
10^{10}
10^{9}
10^{8}
10^{7}
10^{2}
10^{3}
10^{4}
Sequence Length N